Problem Statement

BIS standards are authoritative, but discovering the right IS code from dense PDFs is slow and error-prone.

Search Friction

Small manufacturers often know the product description, not the exact standard number.

Compliance Risk

Wrong or missed standards can delay certification, procurement, QA, and documentation.

Hackathon Goal

Return the correct BIS standards quickly, reproducibly, and without hallucinated codes.

Our project converts a BIS PDF corpus into a fast standards lookup system for MSE-style natural language queries.

Solution Overview

A deterministic hybrid retriever tuned for Indian Standards search

Extracts text and IS-code mentions from `dataset.pdf` using PyMuPDF.

Builds cached search structures for repeatable and fast inference.

Combines BM25 lexical ranking with fuzzy token-set similarity.

Adds query expansion, explicit standard hints, category boosts, and method-standard penalties.

Delivered Artifacts

Judge entry point: `inference.py`

Submission: `data/submission.json`

Demo: `streamlit_app.py`

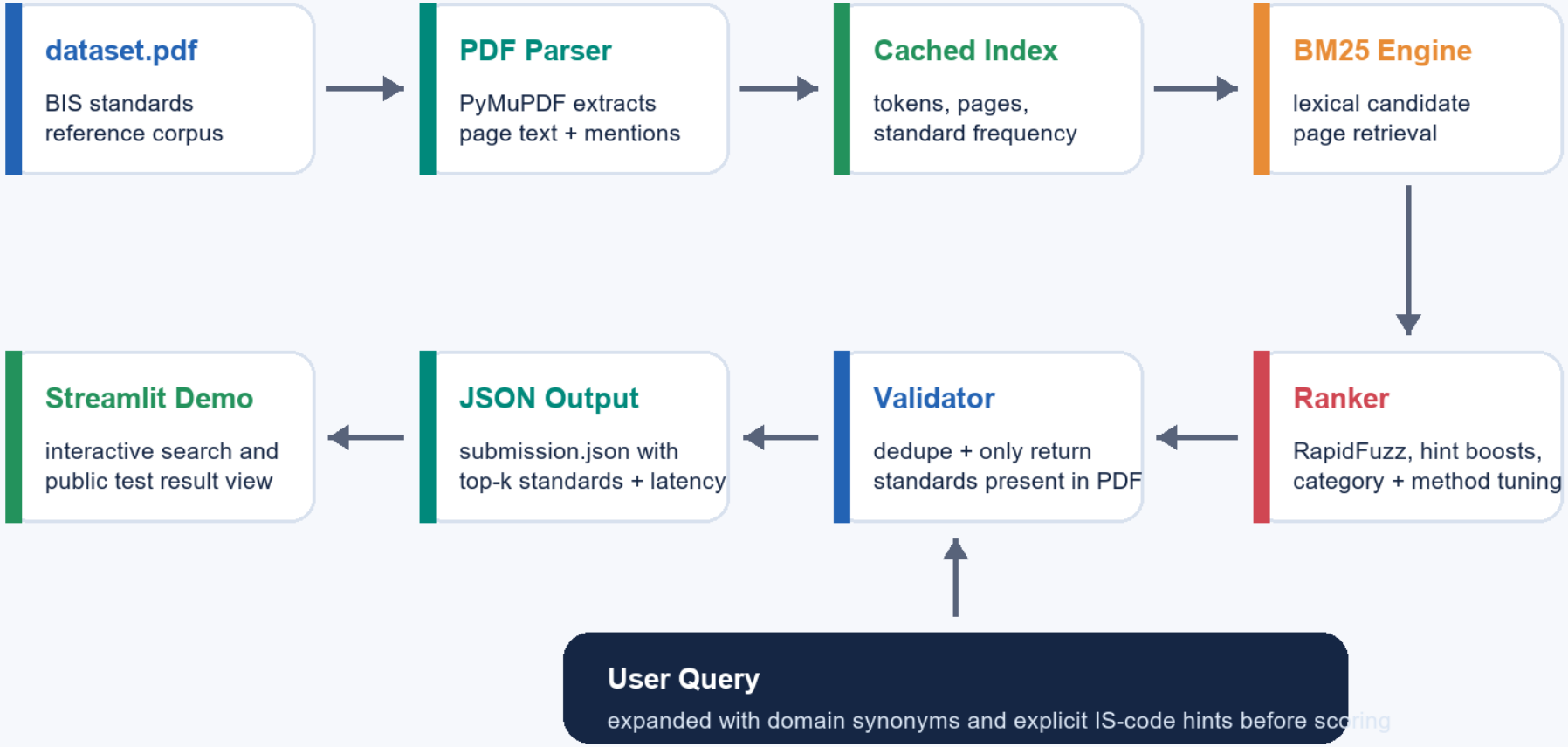
Evaluation: `eval_script.py`

System Architecture

Pipeline generated from the actual project implementation

System Architecture

Deterministic retrieval pipeline for BIS standard discovery



Chunking & Retrieval Strategy

Why the system stays fast while still finding exact IS codes

Chunking

Page-level text extraction preserves local context around standard mentions and avoids over-fragmenting the PDF.

Candidate Retrieval

BM25 ranks top candidate pages from expanded query tokens; the search explores up to 200 pages.

Reranking

RapidFuzz similarity, rarity weighting, year weighting, explicit hints, and category-aware boosts refine the final list.

Standard normalization deduplicates variants such as spacing, parts, and years.

Validation checks returned standards against the source-derived index to avoid hallucinations.

Fallback ranking fills top-k only from standards seen in the PDF corpus.

Demo Highlights

What reviewers can see live

Interactive Search

Streamlit query tab accepts natural language product queries and returns ranked BIS standards with scores/latency.

Public Test View

A second tab loads `data/submission.json`, compares results with expected standards, and marks hits.

Example Queries

33 Grade Ordinary Portland Cement -> IS 269: 1989

Coarse and fine aggregates -> IS 383: 1970

White Portland cement -> IS 8042: 1989

Reproducible Commands

`python inference.py`

`python eval_script.py --results data/submission.json`

`streamlit run streamlit_app.py`

Evaluation Results

Measured on the 10-query public test set

Hit Rate @3

100.00%

Target: >80%

MRR @5

0.8167

Target: >0.7

Avg Latency

0.03 sec

Target: <5 sec

All 10 public queries have at least one expected standard in the top 3 results.

Most product-specific queries rank the target at position 1; harder cement variants still land within top 3.

Cached indexing keeps inference lightweight and deterministic for judge evaluation.

Impact on MSEs

Turning dense standards documents into practical compliance support

Faster Compliance

Teams can move from product description to relevant IS code in milliseconds instead of manual PDF search.

Lower Risk

Validated retrieval reduces missed standards and discourages unsupported answers.

Better Operations

Useful for certification prep, procurement specs, QA checklists, and customer documentation.

The system is especially useful for small teams that need reliable standards discovery without maintaining a full compliance research function.

Team & Acknowledgements

BIS Standards Retriever

Team Members

Hiten Anand
Manavi Bangani

Thank you